



الاختبار النصفى لمقرر التعلم العميق المتقدم (AI4204)

Midterm Exam – Advance Deep Learning (AI4203)

الفصل الدراسي	تاريخ الاختبار	مدة الاختبار
472	الأربعاء 1447/10/20 الموافق 2026/04/08	ساعة

تعليمات الإختبار:

- عدد أسئلة الاختبار ١٢ سؤالاً.
- درجة الاختبار ٣٠ درجة.
- تأكد من كتابة اسمك ورقمك الجامعي ورقم الشعبة في ورقة الإجابة.
- غير مسموح باستخدام أي مذكرات أو مراجع أثناء الاختبار (Closed book exam).
- في حال احتجت إلى مساعدة أو استفسار، ارفع يدك وانتظر المراقب للرد علي ك.
- احرص على متابعة وقت الاختبار لكي تتمكن من الإجابة على جميع الأسئلة
-

Exam Instructions

- The exam consists of 12 questions.
- The total exam score is 30 marks.
- Make sure to clearly write your name, university ID number, and section number on the answer sheet.
- No notes or reference materials are allowed during the exam (closed-book exam).
- At the end of the exam, submit both the question paper and the answer sheet to the invigilator.
- If you need assistance or have a question, raise your hand and wait for the invigilator to assist you.
- Please manage your time carefully to ensure you are able to answer all required questions.

رقم الشعبة	اسم الطالب	الرقم الجامعي
	عزام حسن حسن السدي	444 007018



1. What are the three main parts that make up the Transformer architecture?

- 1- Encoder
- 2- Decoder
- 3- self-attention block

2. What is the primary function of "Positional Encoding" in Transformers?

To add unique value to order of token, so we can extract different meaning of same word if its position different.

تفسير مختلف للمعنى
Positional Encoding

تسجيل الحضور
→ encoding positional information

3. In calculating self-attention, three variables taken from database terminology are used. What are they?

Query (Q): The term that is asking which terms are most relevant

Key (K): A label used to index and match words in attention score labels

Value (V): The stored information retrieved as a result of attention

4. In the self-attention mechanism, what does the Query (Q) represent?

- A) The stored information retrieved as a result of the attention process
- ☒ B) The current word that is "asking" which other words are most relevant to it
- C) A label used to index and match words in the attention lookup table
- D) The scaling factor applied before the SoftMax to stabilize gradients

Processes

5. The SoftMax function in the self-attention equation converts similarity scores into probabilities that sum to 1, producing attention weights for each word.

- ☒ A) True
- B) False



6. What is the key difference between Self-Attention and Masked Self-Attention regarding access to other words?

- A) Self-Attention can only look at previous words, while Masked Self-Attention can look at all words
- ☒ B) Self-Attention allows every word to attend to all other words, while Masked Self-Attention restricts each word to attend only to previous words
- C) Both allow full access to all words, but Masked Self-Attention applies dropout for regularization
- D) Masked Self-Attention attends to future words only, while Self-Attention attends to past words only

7. Why do we transpose the Key matrix (K) when multiplying it with the Query

- A) To normalize the values and prevent large numbers
- ☒ B) To make the matrix dimensions compatible for multiplication
- C) To reduce the size of the matrix for faster computation
- D) To apply positional encoding to the input

8. What is the title of the seminal 2017 paper that first introduced transformer architecture, and what specific task was the model originally designed to perform?

Paper name is "Attention is you need", and the architecture has a several tasks like Text Generation by decoder, sentiment analysis, Named Entity Recognition (NER), natural language understanding (NLU). *Machine Translation كنت من مترجم*

9. In Encoder-Decoder Attention, where do Q, K, V come from?

Q come from masked ~~self~~ attention in decoder block, but K and V come from final hidden state of encoder block.

thes values enter to multi-head attention in decoder and the process name

10. Mention two practical uses for the "Context Aware Embeddings" produced by Encoder-Only Transformers? is Cross attention

1- Sentiment Analysis

2- Named Entity Recognition (NER)

*في نظير انت
لا تحسب*



11. Explain what Word Embedding and Positional Encoding are, and how they are combined to form the input of a Transformer model.

Word Embedding is a word vectors that give meaning to word numerically, and Positional Encoding gives to the word meaning depend on their Positional or order.

After we calculate the values of embedding and Positional Encoding we add them together before attention calculation

12. Compare Encoder-Only and Decoder-Only Transformers. Explain the type of attention used in each, their primary goals, and why ChatGPT is considered a Generative Model.

Aspect	Encoder-Only	Decoder-Only
Attention	Multi head attention	masked attention
Goal	calculation attention score	Restricts each word to attend only to previous
Example tasks	document classification	Text Generation

Why ChatGPT is generative:

Because it's ~~decoder~~ has a decoder architecture that has a masked attention.

~ Good Luck ~



Questions Overview (Marks Distribution)

Q#	Question Type	Marks
Q1	Short Answer	2
Q2	Short Answer	2
Q3	Short Answer	2
Q4	MCQ	2
Q5	True/False	2
Q6	MCQ	2
Q7	MCQ	2
Q8	Short Answer	3
Q9	Short Answer	2
Q10	Short Answer	2
Q11	Essay	4
Q12	Essay	5
Total		30